Midterm review

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Topics covered

PAC learning, VC-dim, growth function, Sauer's Lemma, Rademacher complexity, boosting, bias-variance tradeoff, inductive bias, uniform convergence, Hoeffding

Not on exam: model selection, pseudo-dimension, covering #, chaining,

Inductive bias: what is it, how do we encode it?

PAC learnable: $Z = X \times Y$

A hypothesis class $\mathcal{H}$ is (agnostic) PAC-learnable if $\forall\, \varepsilon, \delta \in (0,1)$,
$\exists\, M$ st. $m \geq M$, $\exists$ algo $A : Z^m \to \mathcal{H}$ st. $\forall$ distributions D over X
and $\forall\, f : X \to Y$ (if agnostic, D is over $X \times Y$)
if $(\mathcal{H}, D, f)$ is realizable (ex: $f \in \mathcal{H}$. Not necessary in agnostic)
then if $S \overset{iid}{\sim} D^m$, w.p. $\geq 1-\delta$,

$$L_{D,f}(A(S)) \leq \varepsilon \qquad L_{D,f}(h) := \mathbb{E}_{x \sim D}\, \ell(h, (x, f(x)))$$

$$L_D(A(S)) \leq \varepsilon + \min_{h \in \mathcal{H}} L_D(h) \quad \text{(agnostic)}$$

$$L_D(h) := \mathbb{E}_{(x,y)\sim D}\, \ell(h, (x,y))$$

ε-representative

$S \subseteq Z^m$ is ε-representative (wrt $Z, \mathcal{H}, \ell, D$) if $\forall h \in \mathcal{H}$,

$$|\hat{L}_S(h) - L_D(h)| \leq \varepsilon$$

Uniform Convergence

$\mathcal{H}$ has the uniform convergence property (wrt Z, ℓ) if $\forall\, \varepsilon, \delta \in (0,1)$,
$\exists\, M$ st $m \geq M \Rightarrow \forall$ prob. dist. D over Z, if $S \overset{iid}{\sim} D^m$,
then w.p. $\geq 1-\delta$, S is ε-representative

Thm

If $\mathcal{H}$ has unif. convergence property, ① $\mathcal{H}$ is agnostic PAC learnable
② $A = ERM_{\mathcal{H}}$ is a valid algo.

proof: $\varepsilon/2$ argument (Lemma 4.2)

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Thm (Cor. 4.6) If $|\mathcal{H}| < \infty$ and output of ℓ is in $[0,1]$ then

$\mathcal{H}$ has uniform convergence property and

$$m_{\mathcal{H}}^{UC}(\varepsilon, \delta) \le \left\lceil \frac{1}{2\varepsilon^2} \log\left(2|\mathcal{H}|/\delta\right) \right\rceil$$

No Free Lunch Thm There's no universal learner.

Let $Y = \{0,1\}$, use $\ell_{0\text{-}1}$, A is any algo. As long as $m \le |X|/2$,

$\exists$ distr. D over $X \times Y$ s.t. if $S \overset{iid}{\sim} D^m$,

1) $\exists$ perfect learner $f: X \to Y$ w/ $L_D(f) = 0$

2) w.p. $\geq 1/7$ (over S), $L_D(A(S)) \geq 1/8$

Know implications

Bias-Variance Tradeoff

$$L_D(h_S) = \underbrace{L_D(h_S) - \min_{h \in \mathcal{H}} L_D(h)}_{\substack{\text{estimation error} \\ \text{(generalization / variance)}}} + \underbrace{\min_{h \in \mathcal{H}} L_D(h)}_{\substack{\text{approx. error, bias} \\ \text{inductive bias}}}$$

Empirical Rademacher Complexity

$$\hat{R}_S(\mathcal{F}) := \mathbb{E}_{\sigma} \sup_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i f(z_i), \quad S = \{z_1, \dots, z_m\}, \quad \sigma_i \overset{iid}{\sim} \text{Unif}\{\pm 1\}$$

or if $A \subseteq \mathbb{R}^m$ (ex: $A = \mathcal{F}(S)$)

$$\hat{R}(A) = \mathbb{E}_{\sigma} \sup_{a \in A} \frac{1}{m} \langle \sigma, a \rangle$$

Rademacher Complexity

$$R_m(\mathcal{F}) = \mathbb{E}_{S \sim D^m} \hat{R}_S(\mathcal{F})$$

Thm (3.3 in Mohri, 26.5 in Shalev-Shwartz)
+3.5

F a family of functions from X to $[0,1]$, then $\forall \delta > 0$, w.p. $\geq 1 - \delta$

$\forall f \in F$

$$\underbrace{\mathbb{E}_{z \sim D} f(z)}_{\substack{L_D(h) \\ \text{True risk}}} \le \underbrace{\frac{1}{m} \sum_{i=1}^{m} f(z_i)}_{\substack{\hat{L}_S(h) \\ \text{Empirical risk}}} + \begin{cases} \cancel{2} R_m(F) + \sqrt{\log(1/\delta)/2m} \\ \cancel{2} \hat{R}_S(F) + 3\sqrt{\log(2/\delta)/2m} \end{cases}$$

eg: $F = \ell \circ \mathcal{H}$
$Y = \{\pm 1\}$

Jensen's Inequality for convex/concave functions

Which way does it go? Take $\sigma \sim$ Rademacher, $\mathbb{E}\sigma = 0$, $\mathbb{E}\sigma^2 = 1$

$$(\mathbb{E}\sigma)^2 \le \mathbb{E}\sigma^2 \quad \text{since } 0 \le 1$$

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Growth Functions

$$\tau_{\mathcal{H}}(m) = \max_{\substack{C \subseteq X \\ |C| = m}} |\mathcal{H}_C| = \max \text{ \# dichotomies given } m \text{ samples}$$

If $|Y| = 2$, $\tau(m) \le 2^m$.

Thm (Cor. 3.8 Mohri), follows from Massart's Lemma

$$R_m(\mathcal{H}) \le \sqrt{\frac{1}{m} 2 \cdot \log(\tau_{\mathcal{H}}(m))} \text{ if } Y = \{\pm 1\}.$$

See also Thm 6.11 in Shalev-Shwartz.

$r = \max_{a \in A} \|a\|$ then if $|A| < \infty$,

$$R(A) \le \frac{r}{m} \cdot \sqrt{2 \log |A|}$$

VC dim

For $|Y| = 2$, $\text{VCdim}(\mathcal{H})$ = maximal size of a set that can be shattered

$= \max\{m : \tau_{\mathcal{H}}(m) = 2^m\}$

No Free Lunch Thm, revisited

$|Y| = 2$, $\text{VCdim}(\mathcal{H}) = \infty \Rightarrow \mathcal{H}$ isn't PAC learnable

Sauer's Lemma

$$\tau_{\mathcal{H}}(m) \le \sum_{i=0}^{d} \binom{m}{i} \quad (d = \text{VCdim}(\mathcal{H}))$$

$$\le \left(\frac{em}{d}\right)^d \text{ if } m \ge d$$

Fund. Thm of PAC learning, $|Y| = 2$, 0-1 loss

T.F.A.E.

1. $\mathcal{H}$ has u.c. property
2. $\mathcal{H}$ is PAC learnable
3. $\mathcal{H}$ is agnostic PAC learnable
4. ERM is a successful (agnostic) PAC learner
5. $\text{VCdim}(\mathcal{H}) < \infty$